MID-TERM EXAMINATION
(Course Name: B. Tech.) (Semester: I)
(October, 2025) OFF LINE mode
For Batches 2024 and 2025

Subject Code: BAS 101	**Subject: Applied Mathematics**
Time : 1 ½Hours	**Maximum Marks :30**

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		**(2.5*4=10)**	**CO Mapping**
	a) Using Cayley-Hamilton Theorem, find A^6 if $A=\begin{bmatrix}2 & 1\\ 5 & -2\end{bmatrix}$.		**CO1**
	b) For what values of x, the rank of matrix $\begin{bmatrix}3-x & 2 & 2\\ 2 & 4-x & 1\\ -2 & -4 & -1-x\end{bmatrix}$ is not equal to 3.		**CO1**
	c) If $x^x\, y^y z^z = k$, then show that $\frac{\partial^2 z}{\partial x \partial y} = -(x\log(ex))^{-1}$ at x=y=z.		**CO2**
	d) Find all the stationary points for the function $f(x, y) = (x^2+y^2)e^{-(x^2+y^2)}$.		**CO2**
	UNIT I		
Q2	**Attempt any two parts**	**(5*2=10)**	
	a) For what values of λ and μ, the system of equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + \lambda z = \mu$ have (i) No solution (ii) A unique Sol (iii) Infinite solutions.		**CO1**
	b) Find the modal matrix for $A=\begin{bmatrix}1 & 1 & 3\\ 1 & 5 & 1\\ 3 & 1 & 1\end{bmatrix}$, if it exists. If not, give reasons.		**CO1**
	c) Are the vectors (1, 2, 4), (2, -1, 3), (0, 1, 2) and (-3, 7, 2) linearly dependent? If so, find the relation between them.		**CO1**
Q3	**Attempt any two parts**	**(5*2=10)**	
	a) If $y = \sin(\log(x^2 + 2x + 1))$, prove that $(1+x^2)y_{n+2} + (2n+1)(1+x)y_{n+1} + (n^2+4)y_n = 0$.		**CO2**
	b) Expand x^y in powers of (x-1) and (y-1) upto the second-degree terms and hence evaluate $(1.1)^{1.02}$.		**CO2**
	c) If $u = x^3y^2 \sin^{-1}\left(\frac{x}{y}\right) + x^{1/3}y^{-1/2}\sin^{-1}\left(\frac{y}{x}\right)$, find $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.		**CO2**